

Map Scale

Scale is the relationship between distance on the map and distance on the ground

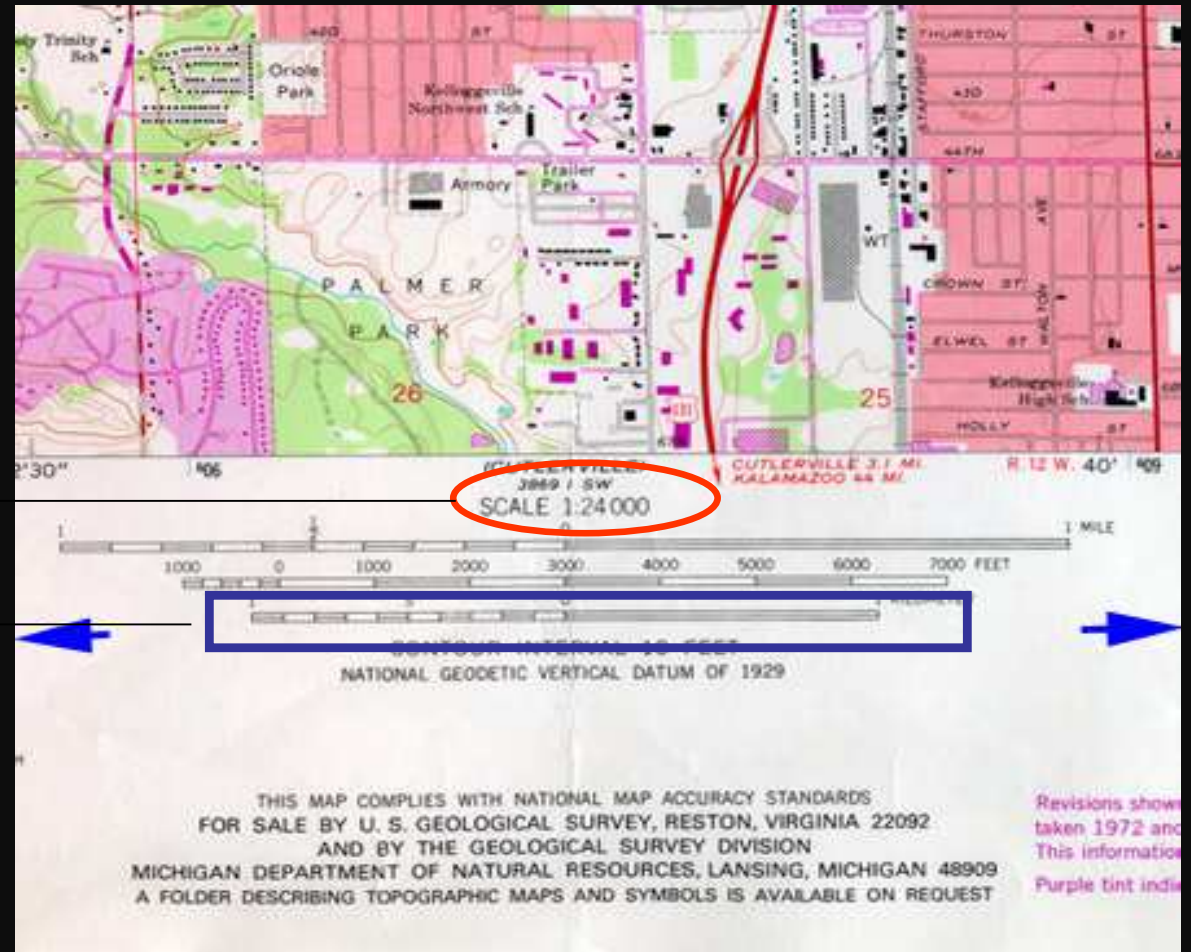
- Word Statement
- Representative Fraction (RF)
- Graphic Scale

Why Scale Imp?

Types

RF

GS



Variation in Scales

Small
→



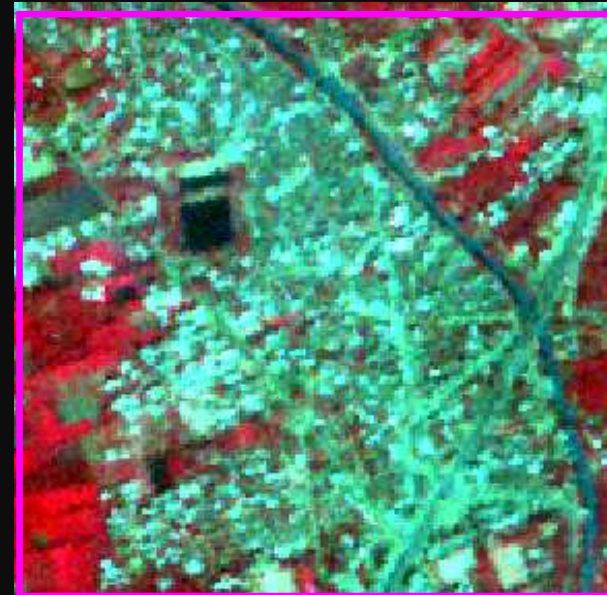
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1:50000



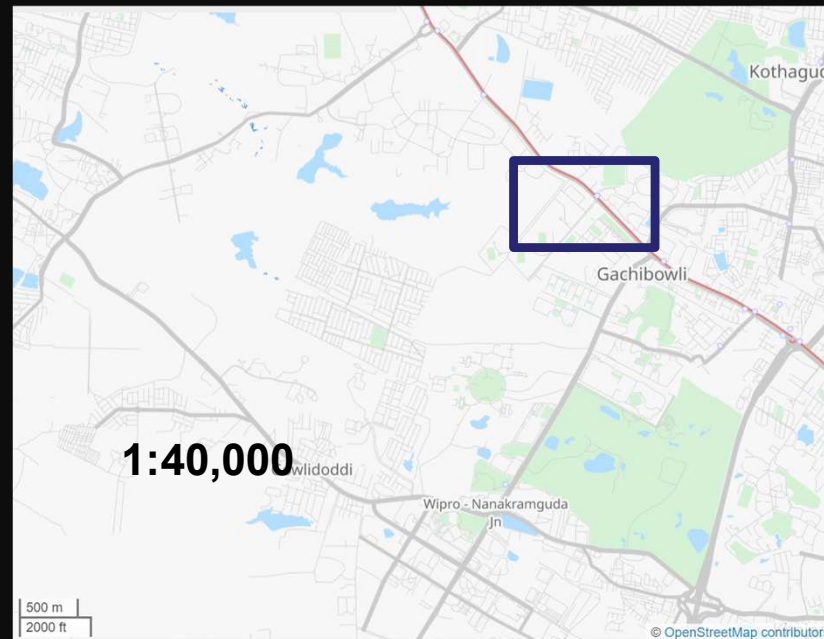
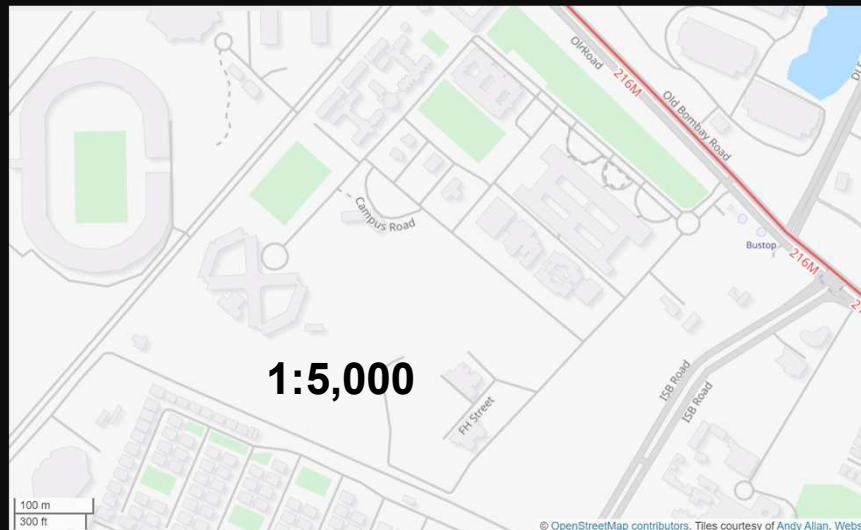
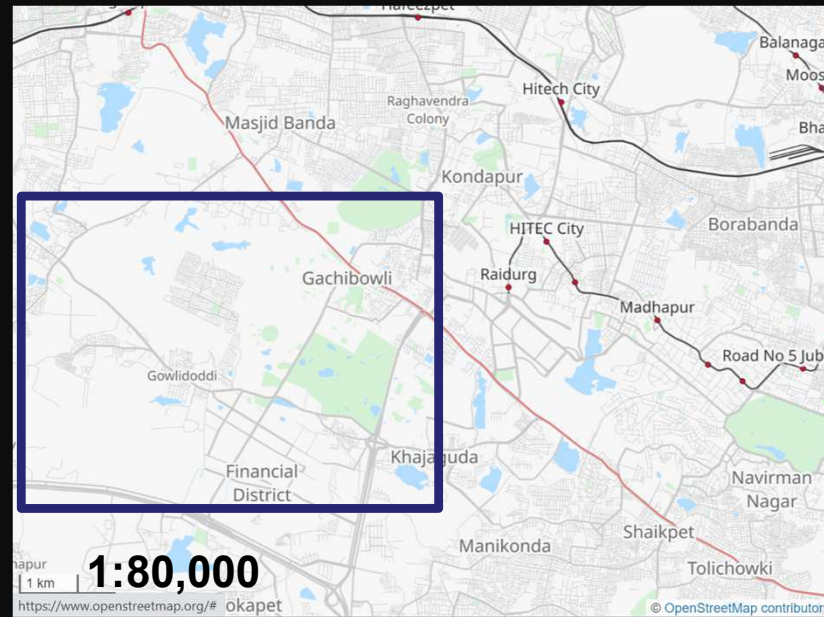
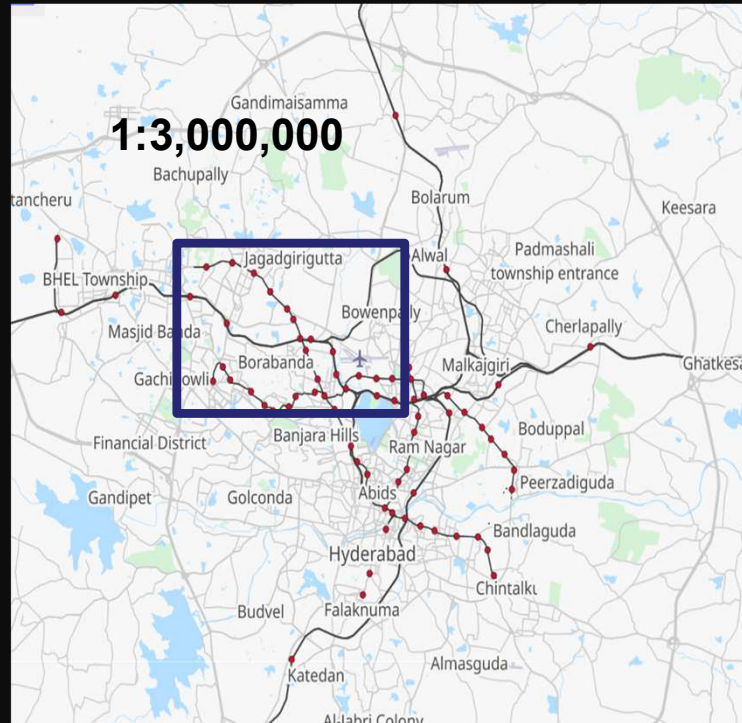
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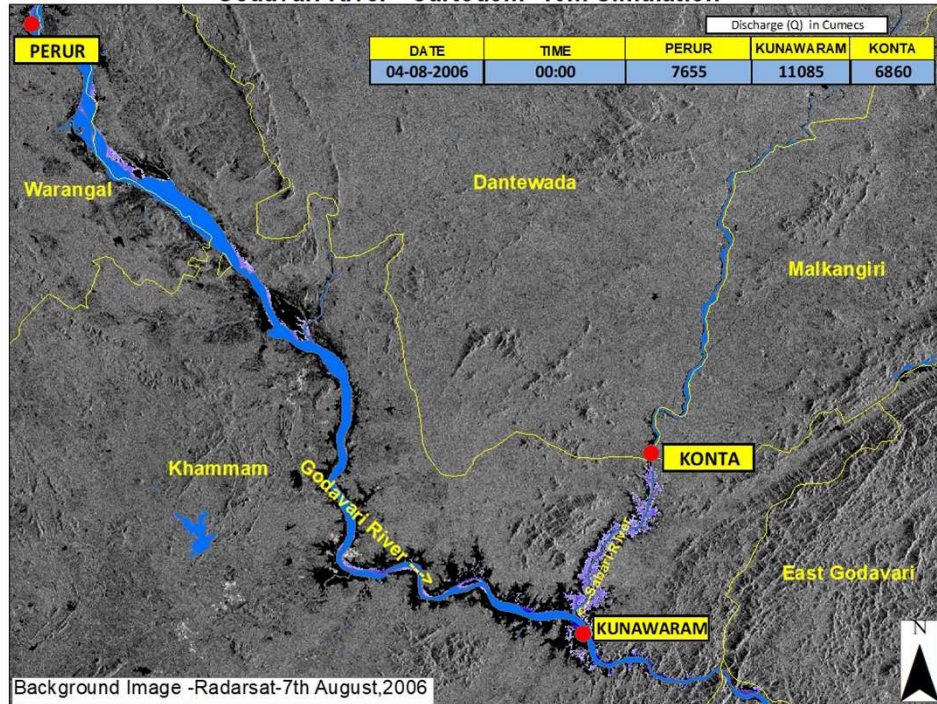
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←
Large

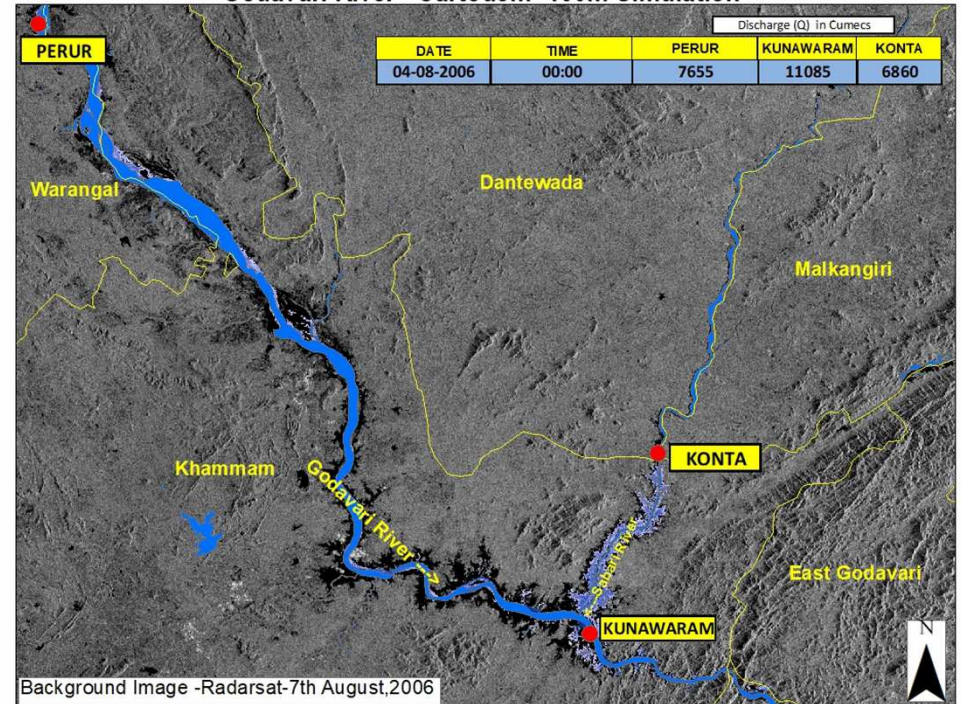
Variation in Scales



Godavari River - Cartodem -10m Simulation



Godavari River - Cartodem -100m Simulation



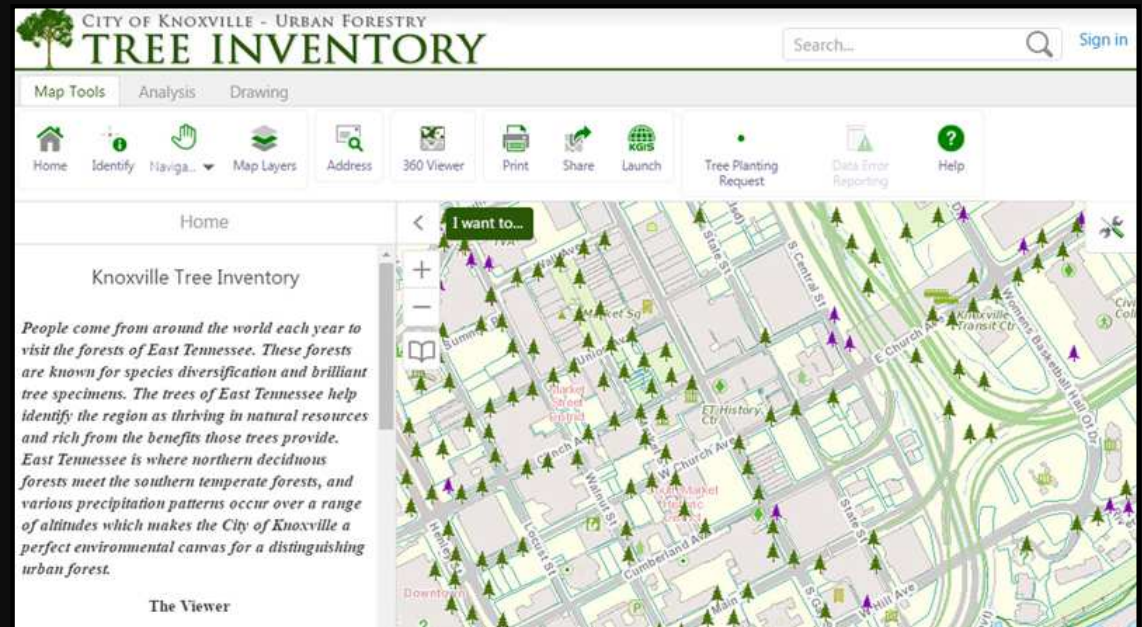
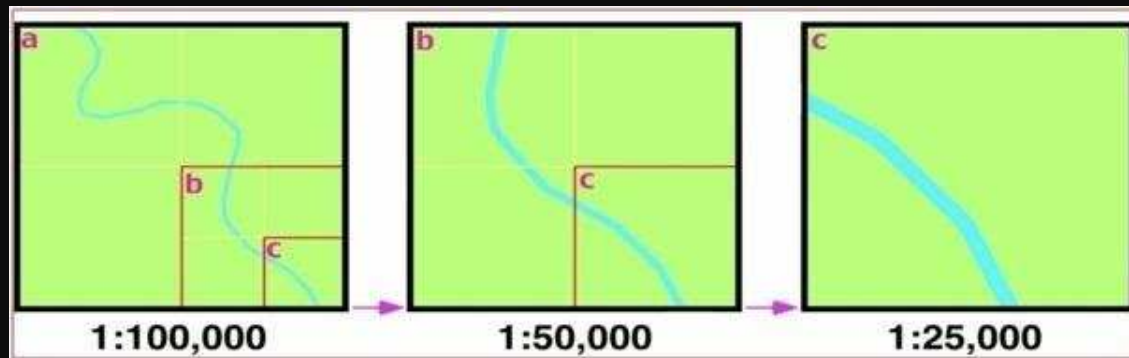
Scale impacts on Simulations

- Slide showing hydrologic modeling outputs (animation) using different resolution DEM models



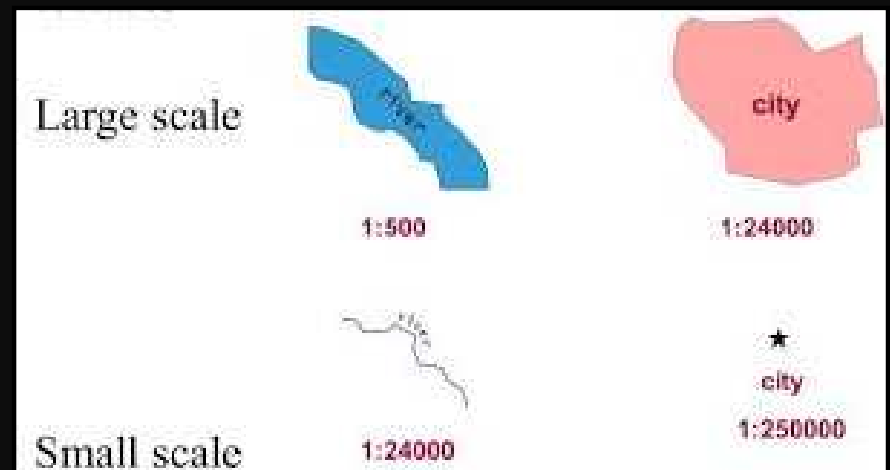
City as point locations

Impact on Line Data



Tree locations in the city

Why Scale Matters?
Analysis, DI, Visualization



What is Map generalization ?

Process of designing the map to reduce the complexities of the real world by strategically reducing ancillary and unnecessary details.

Generalized maps are those that emphasize the most important map elements while still representing the world in the most faithful and recognizable way.

Why

To display accurately the underlying distribution of data.

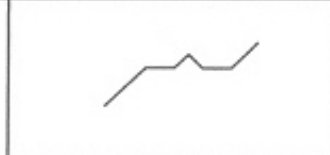
Data is generalized to search and identify for a spatial pattern.

If the data is not properly generalized, the message may be obscured.

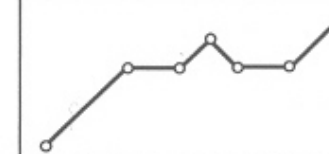
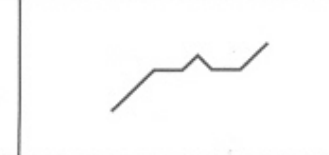
"The process of generalization requires the selection of those features **that are essential** to the map's purpose and the representation of them in a way which is clear and informative".

Methods of Generalization

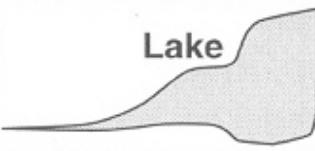
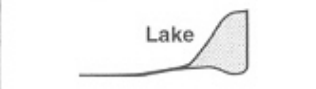
Simplification, for example by weeding out points in the outline of a Polygon to create a simpler shape

Spatial Transformation (Operator)	Representation in	
	Original Map	Generalized Map
Simplification	At Original Map Scale	
		
	At 50% Scale	
		

Smoothing, or the replacement of sharp and complex forms by Smoother ones

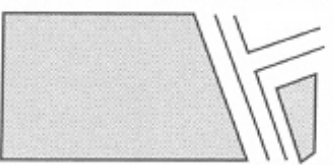
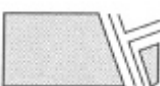
Spatial Transformation (Operator)	Representation in	
	Original Map	Generalized Map
Smoothing	At Original Map Scale	
		
	At 50% Scale	
		

Collapse, or the replacement of an area object by a combination of point and line objects

Spatial Transformation (Operator)	Representation in	
	Original Map	Generalized Map
Collapse	At Original Map Scale	
		
	At 50% Scale	
		

Spatial Transformation (Operator)	Representation in	
	Original Map	Generalized Map
Aggregation	At Original Map Scale	
		
	At 50% Scale	
		

Aggregation, or the replacement of a large number of distinct symbolized objects by a smaller number of new symbols

Spatial Transformation (Operator)	Representation in	
	Original Map	Generalized Map
Amalgamation	At Original Map Scale	
		
	At 50% Scale	
		

Amalgamation, or the replacement of several area objects by a single area object

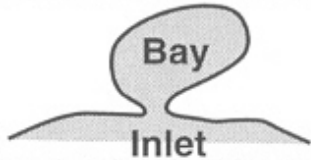
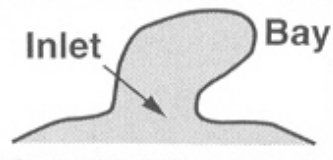
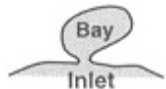
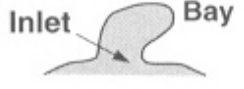
Merging, or the replacement of several line objects by a smaller number of line objects

Spatial Transformation (Operator)	Representation in	
	Original Map	Generalized Map
Merge	At Original Map Scale	
		
	At 50% Scale	
		

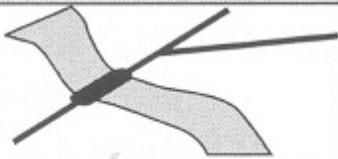
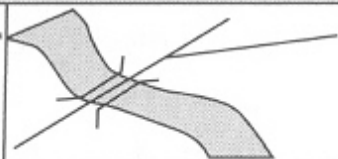
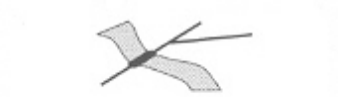
Spatial Transformation (Operator)	Representation in	
	Original Map	Generalized Map
Refinement	At Original Map Scale	
		
	At 50% Scale	
		

Refinement, or the replacement of complex pattern of objects by a selection that preserves the pattern's general form

Exaggeration, or the relative enlargement of an object to preserve its characteristics when these would be lost if the object were shown to scale.

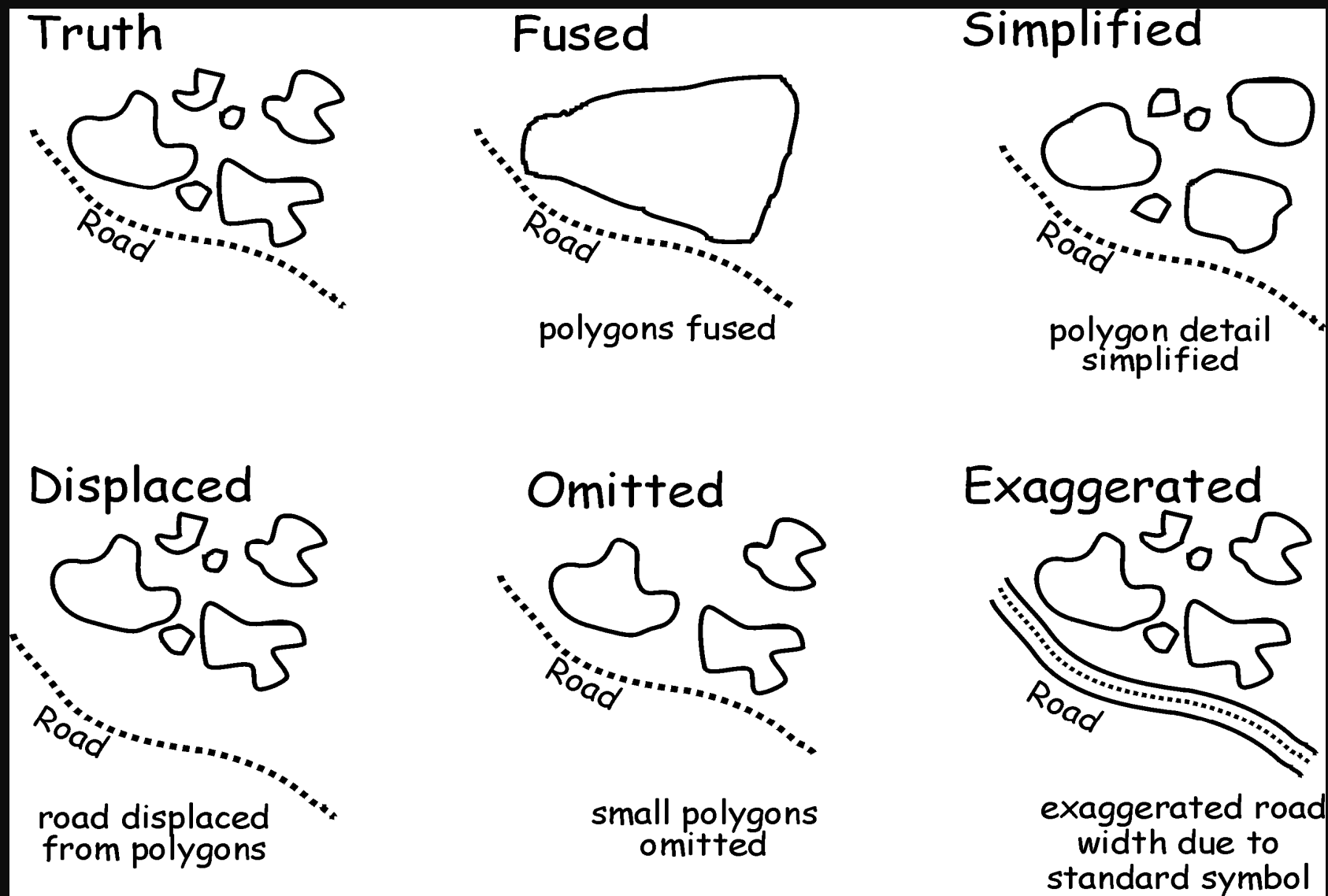
Spatial Transformation (Operator)	Representation in	
	Original Map	Generalized Map
Exaggeration	At Original Map Scale	
		
	At 50% Scale	
		

Enhancement, through the alteration of the physical sizes and shapes of symbols.

Spatial Transformation (Operator)	Representation in	
	Original Map	Generalized Map
Enhancement	At Original Map Scale	
		
	At 50% Scale	
		

Spatial Transformation (Operator)	Representation in	
	Original Map	Generalized Map
Displacement	At Original Map Scale	
		
	At 50% Scale	
		

Displacement, or the moving of objects from their true positions to preserve their visibility and distinctiveness



Generalization of Attributes

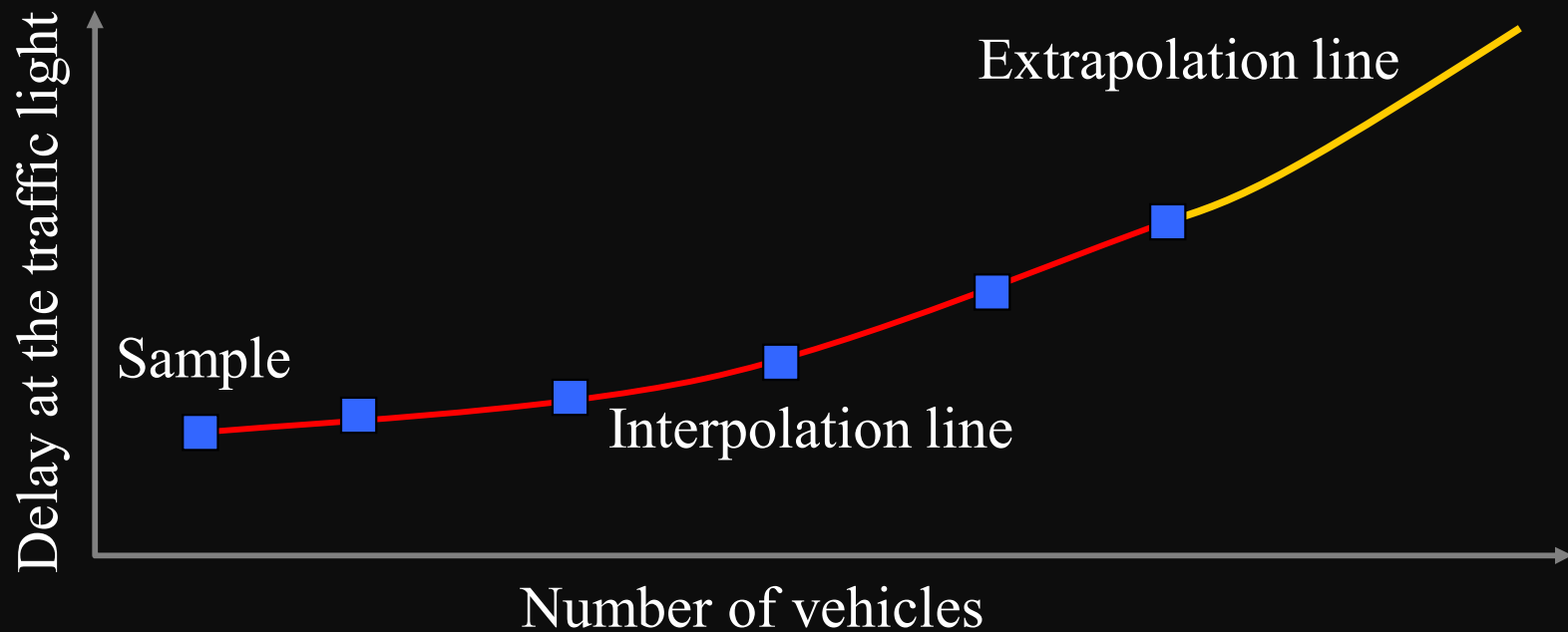
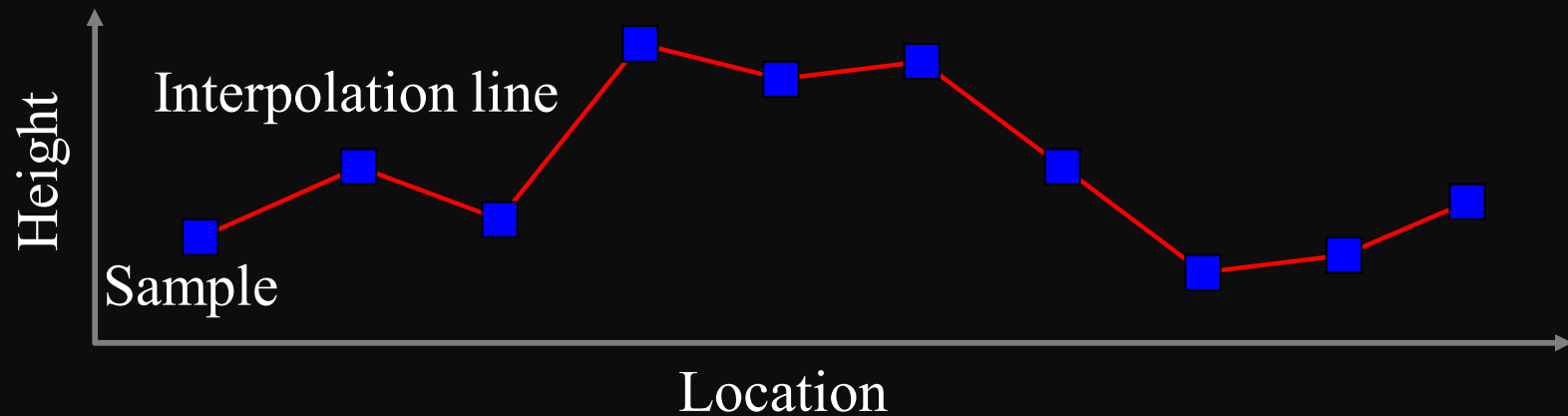
Classification generalization

reclassifies the attributes of objects into Small number of classes

Spatial Inference

- Filling the gaps
 - Sampling shortens the time necessary to collect data.
 - Requires methods to “fill the gaps”.
- Interpolation and extrapolation
 - Data at non-sampled locations can be predicted from sampled locations.
 - **Interpolation:**
 - Predict missing values when bounding values are known.
 - **Extrapolation:**
 - Predict missing values outside the bounding area.
 - Only one side is known.

Spatial Inference: Interpolation and Extrapolation



Querying Spatial Data

Cartographic / Map queries -

Find an ATM, Post Office, etc, Identify MG Road, 44KVA Power line, etc,
Locate a District, Zone, etc

Attribute level queries -

- Which Road is longer – MG Road or RP Road in Secunderabad ?
- Which ATM has more machines?

Spatial queries

- Which is the nearest Hospital ?
- Does CR Rao road intersect a railway line ?

Spatio-Attribute queries

- Find the nearest ATM that has more than 4machines
- Find a theatre that has a Chinese restaurant close to it

Topological queries

- Which plots or survey nos. are going to get affected by road widening?
- Is IIIT Hyderabad in Hyderabad District ?

Example spatial query

“Select all airports in India that belongs to the seismic zone V”

